

D1 Checklist for a hardware project documentation for a BS CS student:

1. Project Overview

- **Project Title:** Clear and concise title reflecting the project's purpose.
- **Summary:** Brief summary of the project, including its goals, objectives, and key features.
- **Introduction:**
 - Background and motivation for the project.
 - Problem statement: Clearly defined problem the project aims to solve.
 - Project scope: Boundaries and limitations of the project.
 - Project goals and objectives: Specific targets to be achieved.

2. Hardware Design and Implementation

- **System Architecture:**
 - High-level block diagram of the system.
 - Detailed description of each component and its function.
- **Circuit Design:**
 - Schematic diagrams: Clear and well-labeled diagrams of the circuit.
 - PCB layout: Physical layout of the circuit components on the PCB.
 - Component selection and justification: Explanation of the choice of components to be purchased.
- **Hardware Implementation:**
 - Assembly process: Step-by-step instructions for assembling the hardware.
 - Hardware Demonstration: Implement a simple hardware circuit and demonstrate its functionality through practical testing.

3. Software Design and Implementation

- **Software Architecture:**
 - High-level overview of the software system.
 - Use Case Diagram: Level 0, Level 1 (Description of the software modules and their interactions.)